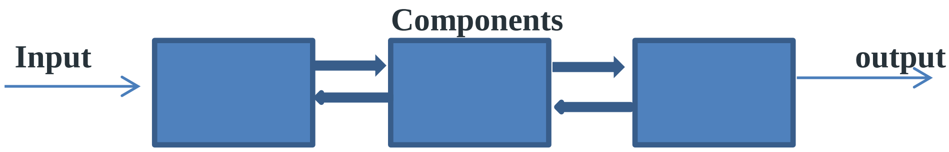
Unit-1

❖ SYSTEM

- The word system is derived from the Greek word “systema” which means a organized relationship among the following unit or component.
- System is collection of component where each component work together to achieve common goal
- A system is an orderly grouping of interdependent components linked together according to a plan to achieve a specific objective.



For example:-

- ✓ A business organization is a system with its components as: Marketing, Manufacture, Sales, Research, Shipping, Accounting, and Administrative All these components work together to **create a profit**.
- ✓ Computer is a system with physical components like keyboard, mouse, scanner, monitor, printers and connecting cables and the basic goal is processing of data.

Characteristics of a System

A system has the following properties –

1.Organization

Organization implies structure and order. It is the arrangement of components that helps to achieve predetermined objectives.

2. Interaction

It is defined by the manner in which the components operate with each other.

For example, in an organization, purchasing department must interact with production department and payroll with personnel department.

3. Interdependence

Interdependence means how the components of a system depend on one another. For proper functioning, the components are coordinated and linked together according to a specified plan. The output of one subsystem is the required by other subsystem as input.

4. Integration

Integration is concerned with how a system components are connected together. It means that the parts of the system work together within the system even if each part performs a unique function.

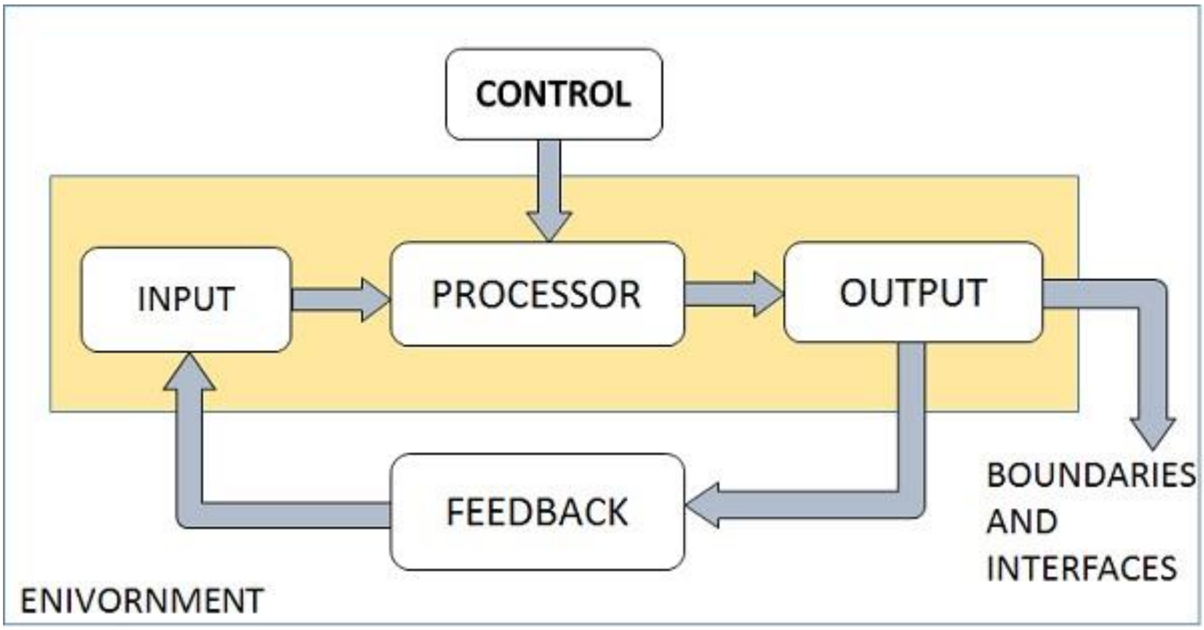
5. Central Objective

The objective of system must be central. It may be real or stated. It is not uncommon for an organization to state an objective and operate to achieve another.

The users must know the main objective of a computer application early in the analysis for a successful design and conversion

Elements of a System

The following diagram shows the elements of a system −



Outputs and Inputs

- The main aim of a system is to produce an output which is useful for its user.
- Inputs are the information that enters into the system for processing.
- Output is the outcome of processing.

Processor(s)

- The processor is the element of a system that involves the actual transformation of input into output.
- It is the operational component of a system. Processors may modify the input either totally or partially, depending on the output specification.

.

Control

- The control element guides the system.
- It is the decision–making subsystem that controls the pattern of activities governing input, processing, and output.
- The behavior of a computer System is controlled by the Operating System and software. In order to keep system in balance, what and how much input is needed is determined by Output Specifications.

Feedback

- Feedback provides the control in a dynamic system.
- Positive feedback is routine in nature that encourages the performance of the system.
- Negative feedback is informational in nature that provides the controller with information for action.

Environment

- The environment is the “super system” within which an organization operates.
- It is the source of external elements that strike on the system.
- It determines how a system must function. For example, vendors and competitors of organization’s environment, may provide constraints that affect the actual performance of the business.

Boundaries and Interface

- A system should be defined by its boundaries. Boundaries are the limits that identify its components, processes, and interrelationship when it interfaces with another system.
- Each system has boundaries that determine its sphere of influence and control.
- The knowledge of the boundaries of a given system is crucial in determining the nature of its interface with other systems for successful design.

Types of Systems

The systems can be divided into the following types −

Physical or Abstract Systems

- **Physical systems** are tangible entities. We can touch and feel them.
- Physical System may be static or dynamic in nature. For example, desks and chairs are the physical parts of computer centre which are static. A programmed computer is a dynamic system in which programs, data, and applications can change according to the user's needs.
- **Abstract systems** are non-physical entities or conceptual that may be formulas, representation or model of a real system.
- For example, an educational system: Our social system.

Open or Closed Systems

- **An open system** must interact with its environment. It receives inputs from and delivers outputs to the outside of the system. For example, an information system which must adapt to the changing environmental conditions.
- **A closed system** does not interact with its environment. It is isolated from environmental influences. A completely closed system is rare in reality.

Adaptive and Non Adaptive System

- **Adaptive System** responds to the change in the environment in a way to improve their performance and to survive. For example, human beings, animals.
- **Non Adaptive** System is the system which does not respond to the environment. For example, machines.

Natural and Manufactured System

- **Natural systems** are created by the nature and there is no interference of human.
- . For example, Solar system, seasonal system.
- **Manufactured System** is the made by the human.
- **It** is also known as man made system
- For example, Rockets, dams, trains.

Deterministic or Probabilistic System

- Deterministic system operates in a predictable manner and the interaction between system components is known with certainty. For example, two molecules of hydrogen and one molecule of oxygen makes water.
- Probabilistic System shows uncertain behavior. The exact output is not known. For example, Weather forecasting, mail delivery.

Social, Human-Machine, Machine System

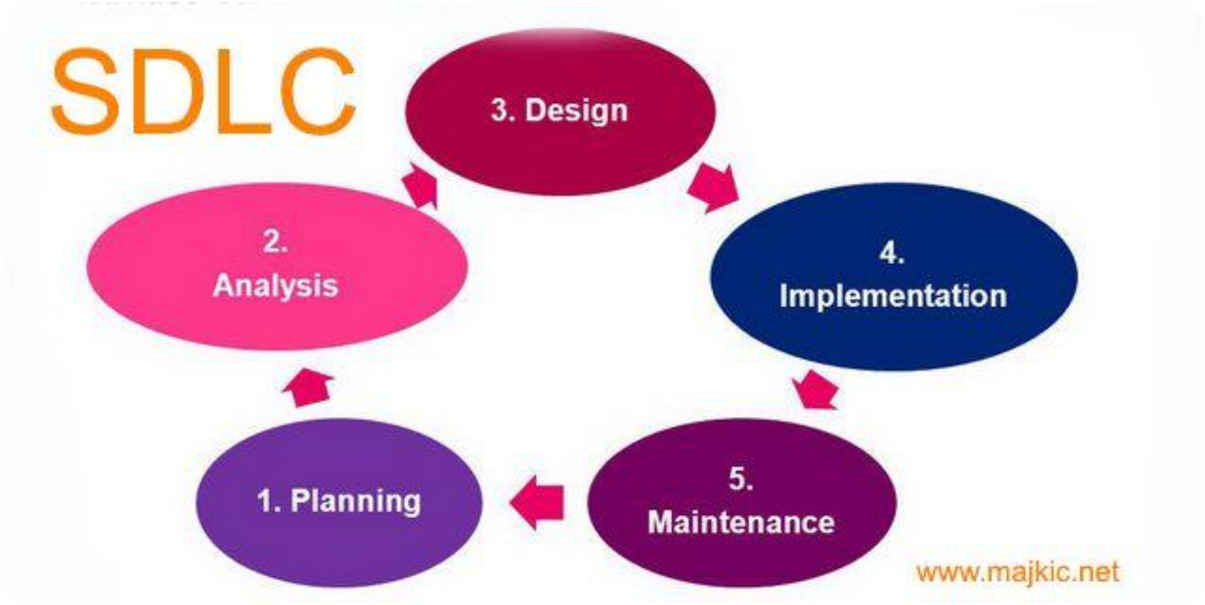
- **Social System** is made up of people. For example, social clubs, societies.
- **In Human-Machine** System, both human and machines are involved to perform a particular task. For example, Computer programming.
- **Machine System** is where human interference is neglected. All the tasks are performed by the machine. For example, an autonomous robot.

Software Development Life Cycle (SDLC)

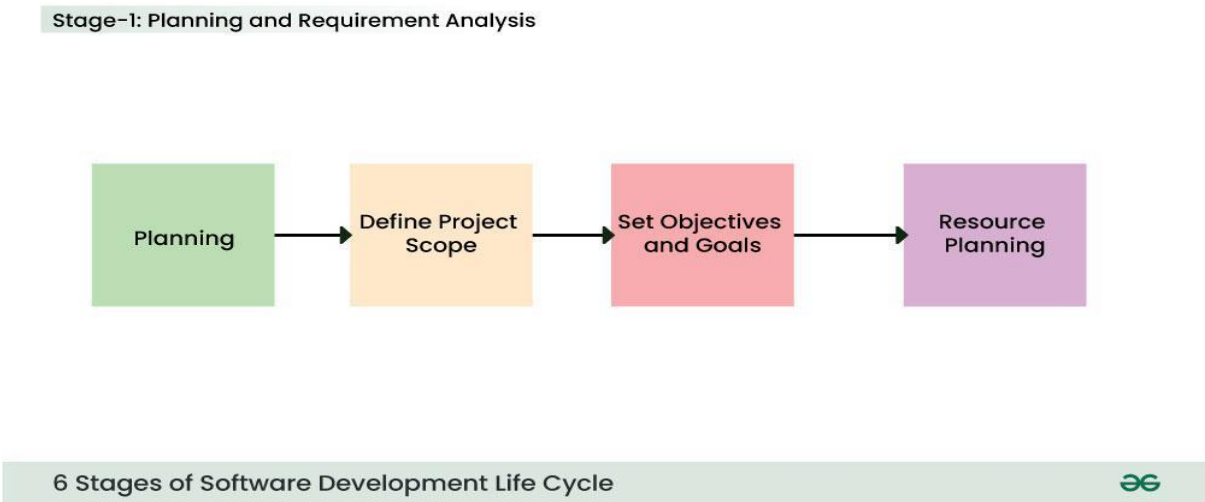
- **SDLC** stands for **Software Development Life Cycle.**
- **Software development life cycle (SDLC) is a structured process that is used to design, develop, and test good-quality software.**
- SDLC, or software development life cycle, is a methodology that defines the entire procedure of software development step-by-step.
- The SDLC is a structured approach that provides a framework for planning, structuring, and controlling the process of developing an information system.
- The goal of the SDLC life cycle model is to deliver high-quality, maintainable software that meets the user’s requirements.

Phase/Stages of the Software Development Life Cycle Model

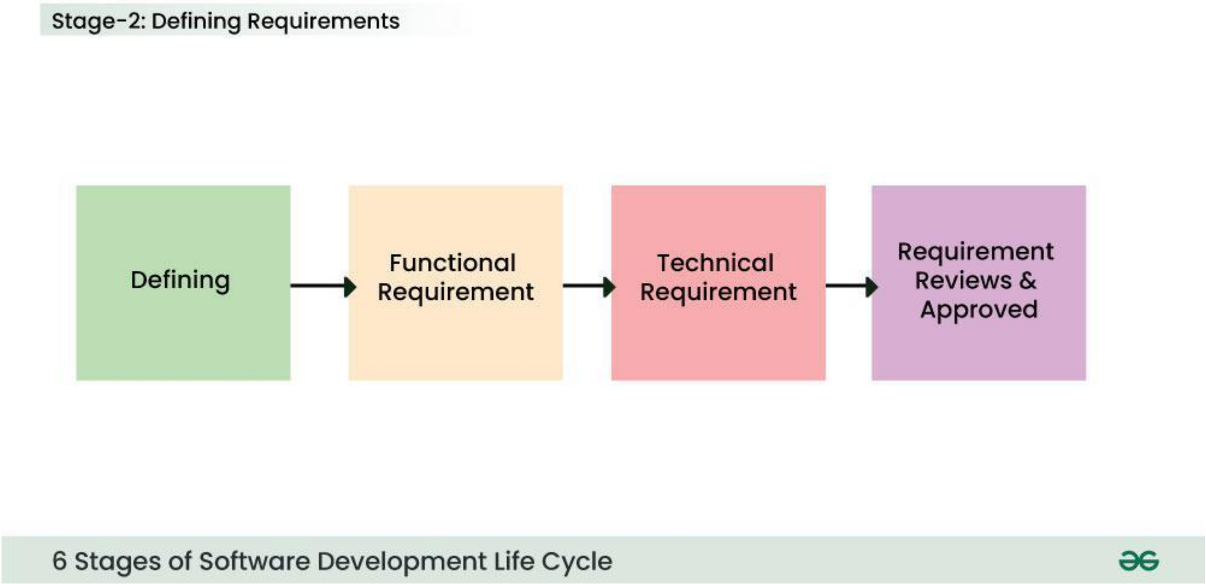
SDLC specifies the task(s) to be performed at various stages by a software engineer or developer.



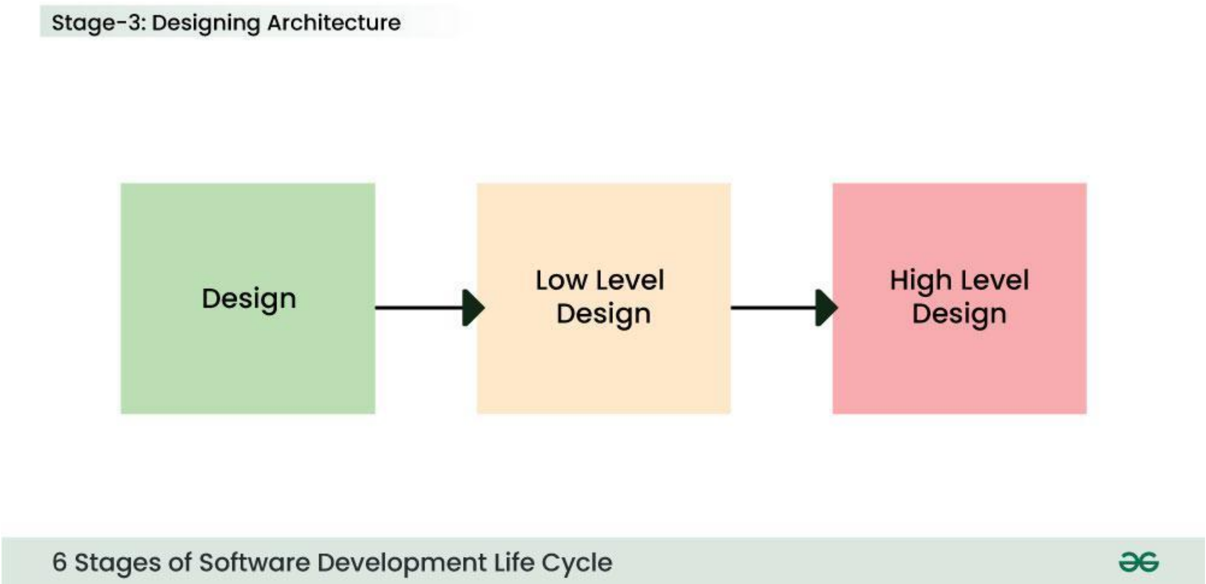
1. **Planning:** This phase involves defining the project, determining its scope, and creating a project plan. In planning phase discusse about set objectives, rule, policy, procedure, cost and budget of any types of system and risk identification.



2. **Analysis:** In this phase, the system requirements are gathered from stakeholders. Analysts study the existing system (if any) and identify the new requirements or improvements needed. This phase is crucial for understanding the user needs and expectations.



3. **Design:** Based on the requirements gathered in the analysis phase, system designers create detailed specifications for the system. This includes architectural design, data structures, interfaces, and other design elements. The goal is to create a blueprint for the development team to follow.



4. **Implementation (Coding):** This phase involves actual coding or programming of the system. Developers use the design specifications to build the system. The implementation phase is where the actual software or application is developed.

- ✓ **Testing:** Once the software is developed, it undergoes testing to ensure that it meets the specified requirements and works as intended. Testing involves identifying and fixing bugs and making sure the software is reliable and error-free.

5. **Maintenance:** After deployment, the system enters the maintenance phase. This involves making updates, fixing bugs, and addressing any issues that arise. Maintenance may also include enhancements or modifications to the system to meet changing requirements. We need to up to date system according to super environment for more survival chance.

Role of System Analyst

✓ System Analysts:-

- ✓ System analysts are a person or individual who use system analysis and design technique to solve organization and business problem.
- ✓ System analyst analysis, design and implement system to fulfil organisation need.
- ✓ System analyst conduct the study identifies requirements and determine the procedures to achieve system objectives.
- ✓ System analyst is responsible for starting to end of any types of system.

Role of system analysts:-

1. **Problem definition**

   The most important and difficult task of an analyst is to understand the organization's requirement's information. It includes interviewing users finding out what information is the year using in the current system.
2. **Setting priorities among requirement**

   In the organization there are many types of users, each user has different types of information needs. It may not possible to satisfy the requirements of everyone due to limited availability of resources so it is necessary to give priority. The priorities are set on the basis of urgency and importance of user's need
3. **Gathered data and fact**

   For gathering data or facts, written documents are important because these documents represent the formal information flow in the system. The analyst studies documents such as input forms, output records, invoices etc. to understand how data are passed and used in the present system. System analysts may also collect data by using different tool interviewing, observation, survey, questionaries' etc. for solving system problems.

4. **Analysis and evaluate:-**

   After collecting the data, system analysts need to analysis of data for the solving of problem and taking decision. Analysis and evaluating very important step in any system for solving problem and taking decision.

5. **Problem solving:-**

   The system analyst helps users to solve problems. In that role he must understand the problem and suggest solutions**.**

6. **Drawing specification:-**

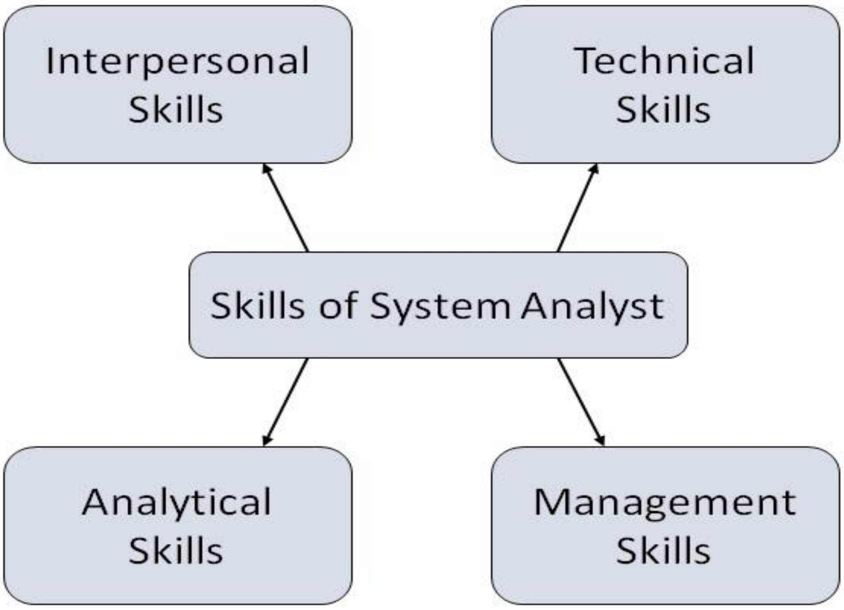
   After solving the problem of user, system analysts make some specification to improve the existing system**.**

7. **Designing system:** Once the specifications are accepted by the management the analyst gets on to the design of the system. The analyst must be aware of the latest design tools for the system design so analyst also knows as architect.

8. **Evaluating system:** An analyst must critically test the performance of the designed system with specifications after it has been in use for a reasonable period of time.

Attributes of a Systems Analyst

The following figure shows the attributes a systems analyst should possess −



Interpersonal Skills

- Interface with users and programmer.

- Facilitate groups and lead smaller teams.
- Managing expectations.
- Good understanding, communication, selling and teaching abilities.
- Motivator having the confidence to solve queries.

Analytical Skills

- System study and organizational knowledge
- Problem identification, problem analysis, and problem solving
- Sound common-sense
- Ability to access trade-off
- Curiosity to learn about new organization

Management Skills

- Understand users jargon and practices.
- Resource & project management.
- Change & risk management.
- Understand the management functions thoroughly.

Technical Skills

- Knowledge of computers and software.
- Keep abreast of modern development.
- Know of system design tools.
- Breadth knowledge about new technologies.

Fact Finding/Information gathering

- ✓ Information gathering is a crucial step in various fields, including business, cyber security, system development, research, and more.
- ✓ It involves collecting relevant data and details to understand a particular subject, solve a problem, or make informed decisions.
- ✓ There are various methods and tools available for information gathering, depending on the context and the nature of the information needed.

Fact finding/Information Gathering Techniques

There are various information gathering techniques –

- ✓ **Interviewing**

- ✓ Systems analyst collects information from individuals or groups by interviewing.
- ✓ The analyst can be formal, legalistic, play politics, or be informal; as the success of an interview depends on the skill of analyst as interviewer.

It can be done in two ways −

- **Unstructured Interview** − The system analyst conducts question-answer session to acquire basic information of the system.

- **Structured Interview** − It has standard questions which user need to respond in either close (objective) or open (descriptive) format.

Advantages of Interviewing

- This method is frequently the best source of gathering qualitative information.
- It is useful for them, who do not communicate effectively in writing or who may not have the time to complete questionnaire.
- Information can easily be validated and cross checked immediately.
- It can handle the complex subjects.
- It is easy to discover key problem by seeking opinions.

✓ Questionnaires

✓ This method is used by analyst to gather information about various issues of system from large number of persons.

There are two types of questionnaires −

- **Open-ended Questionnaires** − It consists of questions that can be easily and correctly interpreted. They can explore a problem and lead to a specific direction of answer.
- **Closed-ended Questionnaires** − It consists of questions that are used when the systems analyst effectively lists all possible responses, which are mutually exclusive.

Advantages of questionnaires

- It is very effective in surveying interests, attitudes, feelings, and beliefs of users which are not co-located.
- It is useful in situation to know what proportion of a given group approves or disapproves of a particular feature of the proposed system.
- It is useful to determine the overall opinion before giving any specific direction to the system project.
- It is more reliable and provides high confidentiality of honest responses.

✓ Review of Records, Procedures, and Forms

✓ Review of existing records, procedures, and forms helps to seek insight into a system which describes the current system capabilities, its operations, or activities.

Advantages

- It helps user to gain some knowledge about the organization or operations by themselves before they impose upon others.
- It can help an analyst to understand the system in terms of the operations that must be supported.
- It describes the problem, its affected parts, and the proposed solution.

- ✓ Observation

- ✓ This is a method of gathering information by noticing and observing the people, events, and objects. The analyst visits the organization to observe the working of current system and understands the requirements of the system.

Advantages

- It is a direct method for gleaning information.
- It is useful in situation where authenticity of data collected is in question or when complexity of certain aspects of system prevents clear explanation by end-users.
- It produces more accurate and reliable data.
- It produces all the aspect of documentation that are incomplete and outdated.

- ✓ Joint Application Development (JAD)

- ✓ It is a new technique developed by IBM which brings owners, users, analysts, designers, and builders to define and design the system using organized and intensive workshops.
- ✓ JAD trained analyst act as facilitator for workshop who has some specialized skills.

Advantages of JAD

- It saves time and cost by replacing months of traditional interviews and follow-up meetings.
- It is useful in organizational culture which supports joint problem solving.
- Fosters formal relationships among multiple levels of employees.
- It can lead to development of design creatively.
- It Allows rapid development and improves ownership of information system.

- ✓ Secondary Research or Background Reading

    - ✓ This method is widely used for information gathering by accessing the gleaned information. It includes any previously gathered information used by the marketer from any internal or external source.

Advantages

- It is more openly accessed with the availability of internet.
- It provides valuable information with low cost and time.
- It act as forerunner to primary research and aligns the focus of primary research.
- It is used by the researcher to conclude if the research is worth it as it is available with procedures used and issues in collecting them.

Fact analysis

- ✓ Fact analysis is a systematic process of examining and evaluating factual information to derive insights, draw conclusions, or make informed decisions.

- ✓ This analytical approach involves organizing, reviewing, and interpreting data or facts to gain a deeper understanding of a situation, problem, or topic.
- ✓ Fact analysis is commonly used in various fields, including business, research, intelligence, law, and more.
- ✓ Fact analysis is an integral part of evidence-based decision-making and problem-solving.
- ✓ It helps in transforming raw data into meaningful information and contributes to a more informed understanding of complex situations. Whether in business, research, or any other domain, fact analysis enhances the ability to make well-founded decisions and develop effective strategies

Here are the key steps involved in fact analysis:

- ✓ **Identification of Relevant Facts:**

- ✓ Begin by identifying and gathering all pertinent facts related to the subject under investigation. This may involve reviewing documents, reports, interviews, surveys, or any other sources of information.

- ✓ **Organization and Documentation:**

- ✓ Systematically organize the collected facts for easy reference and analysis. Documentation may include creating a structured database, using spread sheets, or developing a timeline to chronologically represent events.

- ✓ **Validation of Facts:**

- ✓ Ensure the accuracy and reliability of the collected facts.
- ✓ Cross-check information from multiple sources to validate its authenticity and eliminate any potential bias or misinformation.

- ✓ Some of **the tools** available for data organization and analysis are input/output analysis, decision tables, and structure charts.

1. Input/ Output Analysis

Input/output analysis identifies the elements that are related to the inputs and output of a given system. Flow charts and data flow diagram are excellent tools for input/output analysis.

2. Decision Table:-

This table described the data flow within the system. It is used when complex decision logic can't be represented clearly in the flow chart as a documenting tool they provide a simpler form of data analysis then the flow chart. It is easy to follow communication devices between technical & non-technical person:

3. Structured Chart:

- ✓ Describe functions and sub-functions of each part of system(in more detail than a DFD).
- ✓ Show relationships between common and unique modules of a computer program.
- ✓ **Hierarchical, modular structure**
- ✓ Each layer in a program performs specific activities
- ✓ Each module performs a specific function

Introduction to System Planning and Initial Investigation:

- ✓ System planning and initial investigation are fundamental phases in the System Development Life Cycle (SDLC).
- ✓ During this stage, organizations identify the need for a new information system or modifications to an existing one.
- ✓ The goal is to define the scope of the project, assess its feasibility, and establish a foundation for subsequent phases of system development.

System Planning:

- ✓ System planning involves determining the overall goals and objectives of the proposed system.
- ✓ It includes identifying the problems or opportunities that the system is expected to address and outlining the strategic vision for the project.

Key Activities System Planning:

- ✓ Defining the purpose and scope of the system.
- ✓ Identifying stakeholders and their expectations.
- ✓ Outlining high-level goals and objectives.
- ✓ Establishing the strategic alignment of the system with organizational goals.

Initial Investigation:

- ✓ Initial investigation, also known as system investigation or preliminary analysis, is the process of gathering and evaluating information about the existing system (if any) and the potential requirements for the new system.

Key Activities Initial Investigation:

- ✓ Conducting interviews with stakeholders to understand their needs.
- ✓ Analysing the current business processes and information flow.
- ✓ Identifying problem areas or opportunities for improvement.
- ✓ Assessing the technical, operational, and economic feasibility of the proposed system.

Bases for Planning in System Analysis:

- ✓ Organizations initiate system planning based on identified business needs.
- ✓ Aligning system planning with the organization's strategic objectives ensures that the proposed system supports the broader goals of the business. Market trends, customer demands, and competitive pressures can drive system planning.

- ✓ Advances in technology may present opportunities for system improvements.
- ✓ Input from stakeholders, including employees and customers, can serve as a basis for planning. User feedback, suggestions for improvement, or identified pain points in existing systems may trigger the need for change.
- ✓ The growth or restructuring of an organization may require adjustments to existing systems or the planning of new systems to accommodate changes in scale, structure, or business processes.

Source of project request:-

- ✓ A project is defined as a sequence of tasks that must be completed to attain a certain outcome.

- ✓ Project requests can come from various sources, depending on the nature of the project and the industry.

Here are some common sources of project requests:

- ❖ **Internal Stakeholders**:

- ✓ **Management:** Requests may come from upper management or department heads who identify the need for a specific project to achieve organizational goals.
- ✓ **Employees**: Team members may propose projects based on their observations and experiences in day-to-day operations.

- ❖ **External Customers/Clients:**

- ✓ **Client Requests:** External clients or customers may directly request specific projects or modifications to existing products/services.
- ✓ **Market Feedback**: Feedback from the market can trigger new project ideas or improvements to existing products.

- ❖ **Regulatory Changes:**

- ✓ Legal/Regulatory Requirements: Changes in laws and regulations may necessitate projects to ensure compliance and avoid legal issues.

- ❖ **Technology Changes:**

- ✓ **Advancements in Technology**: Changes in technology can prompt projects to upgrade systems, adopt new tools, or implement innovative solutions.

- ❖ **Strategic Planning:**

- ✓ Strategic Initiatives: Projects may be driven by the strategic goals and initiatives outlined in the organization's strategic plan.

- ❖ **Problem Identification:**

- ✓ **Issue Resolution**: Problems or challenges within the organization or its processes may lead to project requests aimed at finding solutions.

- ❖ **Customer Feedback:**

- ✓ **Customer Surveys:** Feedback from customer surveys can highlight areas for improvement and lead to new projects.
- ✓ **Customer Support:** Issues raised by customer support teams may result in projects to address common customer concerns.